nThe Gram-Schmidt Process

The Gram-Schmidt Process is an algorithm used to convert any basis for a subspace W of $\mathbb{R}^n$ into a new orthogonal basis for W. This orthogonal basis can then be normalized, if desired, to get an orthonormal basis for W. Since every nonzero subspace W has a basis to which the Gram-Schmidt Process can be applied, this means that every nonzero subspace of $\mathbb{R}^n$ has an orthonormal basis.

It is important to think about what the algorithm is doing: that makes it much easier to remember and use the algorithm. Therefore this presentation, while doing the very same thing as in the text, tries harder than the text to separate the idea from the calculations The idea behind Gram-Schmidt Process is contained in the following Lemma.

Lemma Suppose $\{\boldsymbol{v}_1, ..., \boldsymbol{v}_i\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$ and $W = \text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i\}$.

Pick a vector $\boldsymbol{x} \notin W$ and write $\boldsymbol{x} = \widehat{\boldsymbol{x}} + \boldsymbol{v}$ where $\widehat{\boldsymbol{x}} = \text{proj}_W \boldsymbol{x} \in W$
and $\boldsymbol{v} \in W^\perp$.

Then

i) $\text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{x}\} = \text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{v}\}$, and
ii) the enlarged set $\{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{v}\}$ is orthogonal

NOTE: in the Orthogonal Decomposition Theorem, we wrote $\boldsymbol{x} = \widehat{\boldsymbol{x}} + \boldsymbol{z}$ where $\boldsymbol{z} \in W^\perp$. Here, I'm using the letter $\boldsymbol{v}$ (rather than the letter $\boldsymbol{z}$) to denote the orthogonal component of $\boldsymbol{x}$. In this context, I think that helps the discussion to read more smoothly.

Informally, what does the Lemma say?

The $\boldsymbol{v}_i$'s are orthogonal to each other, so $\{\boldsymbol{v}_1, ..., \boldsymbol{v}_i\}$ is a linearly independent set and a basis for W. Then we enlarge W to $\text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{x}\}$ by adding to the basis a new vector $\boldsymbol{x}$ from "outside W." Since $\boldsymbol{x}$ is not a linear combination of $\boldsymbol{v}_1, ., , \boldsymbol{v}_n$, $\{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{x}\}$ is linearly independent and is a basis for the enlarged subspace $\text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{x}\}$.

But "the $\widehat{\boldsymbol{x}}$ part" of $\boldsymbol{x}$ was already in W before we enlarged; it is $\boldsymbol{v}$, the orthogonal component of $\boldsymbol{x}$ (from $W^\perp$) that actually enlarges W. Just adding the orthogonal component of $\boldsymbol{x}$ to $\{\boldsymbol{v}_1, ..., \boldsymbol{v}_i\}$ has the same effect as adding "the whole vector" $\boldsymbol{x}$ to $\{\boldsymbol{v}_1, ..., \boldsymbol{v}_i\}$, and since $\boldsymbol{v} \in W^\perp$, the new set $\{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{v}\}$ is an orthogonal basis for the enlarged subspace.

Proof i) To show that $\text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{x}\} = \text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{v}\}$, we need to show that

a) every vector $\boldsymbol{w}$ in $\text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{x}\}$ is also in $\text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{v}\}$, and
b) every vector $\boldsymbol{w}$ in $\text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{v}\}$ is also in $\text{Span}\ \{\boldsymbol{v}_1, ..., \boldsymbol{v}_i, \boldsymbol{x}\}$

For short, let's use the notation $\text{lc}(\boldsymbol{v}_1, ..., \boldsymbol{v}_i)$ to mean "a linear combination of $\boldsymbol{v}_1, ..., \boldsymbol{v}_i$." With this shorthand, for example, "$\text{lc}(\boldsymbol{v}_1, ..., \boldsymbol{v}_i) + \text{lc}(\boldsymbol{v}_1, ..., \boldsymbol{v}_i) = \text{lc}(\boldsymbol{v}_1, ..., \boldsymbol{v}_i)$" (*why*?)

Since $\widehat{\boldsymbol{x}}$ is in W, we know that $\widehat{\boldsymbol{x}} = \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i})$,

For a) If $\boldsymbol{w}$ is in Span $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}, \boldsymbol{x}\}$, then

$$\begin{aligned}
\boldsymbol{w} &= \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) && + \; c\boldsymbol{x} \\
&= \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) && + \; c(\,\widehat{\boldsymbol{x}} + \boldsymbol{v}\,) \\
&= \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) && + \; c\widehat{\boldsymbol{x}} + c\boldsymbol{v} \\
&= \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) && + \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) + c\boldsymbol{v} \\
&= \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) && + c\boldsymbol{v}, \text{ \underline{so}}
\end{aligned}$$

$\boldsymbol{w}$ is in Span $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}, \boldsymbol{v}\}$.

For b) If $\boldsymbol{w}$ is in Span $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}, \boldsymbol{v}\}$, then

$$\begin{aligned}
\boldsymbol{w} &= \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) \; + c\boldsymbol{v} \\
&= \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) && + c(\boldsymbol{x} - \widehat{\boldsymbol{x}}) \\
&= \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) && + c\boldsymbol{x} - \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) \\
&= \mathrm{lc}(\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}) && + c\boldsymbol{x}, \text{ \underline{so}}
\end{aligned}$$

$\boldsymbol{w}$ is in Span $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}, \boldsymbol{x}\}$

ii) $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}\}$ is an orthogonal set and $\boldsymbol{v} \perp$ (each $\boldsymbol{v_i}$) because $\boldsymbol{v} \in W^{\perp}$. So $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_i}, \boldsymbol{v}\}$ is an orthogonal set. ●

<u>The Gram-Schmidt Process</u> (GSP) If you understand the preceding lemma, the idea behind the Gram-Schmidt Process is very easy. We want to <u>convert</u> an basis $\{\boldsymbol{x_1}, \ldots, \boldsymbol{x_p}\}$ for W into an <u>orthogonal</u> basis $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$. We build the orthogonal basis by replacing each vector $\boldsymbol{x_i}$ with a vector $\boldsymbol{v}_i$. We do this "from the ground up," one $\boldsymbol{x_i}$ at a time, starting with $\boldsymbol{x_1}$.

Step 1. To start, let $\boldsymbol{v_1} = \boldsymbol{x_1}$, and call $W_1 = \text{Span}\,\{\boldsymbol{x_1}\} = \text{Span}\,\{\boldsymbol{v_1}\}$

Step 2: Now enlarge W_1: let $W_2 = \text{Span}\,\{\boldsymbol{x_1}, \boldsymbol{x_2}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{x_2}\}$ (*since* $\boldsymbol{v_1} = \boldsymbol{x_1}$).
Decompose $\boldsymbol{x_2} = \widehat{\boldsymbol{x}}_{\boldsymbol{2}} + \boldsymbol{v_2}$, where $\widehat{\boldsymbol{x}}_{\boldsymbol{2}} = \text{proj}_{W_1}\boldsymbol{x_2}$ and $\boldsymbol{v_2} \in W_1^{\perp}$.
But we don't need "the whole vector $\boldsymbol{x_2}$" to get W_2. According to the Lemma,
$W_2 = \text{Span}\,\{\boldsymbol{x_1}, \boldsymbol{x_2}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{x_2}\} = \text{Span}\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$,
and, by the Lemma, $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is an orthogonal set.

Of course, we already know a formula for $\boldsymbol{v_2}$: since $W_1 = \text{Span}\{\boldsymbol{v_1}\}$,

$$\boldsymbol{v_2} = \boldsymbol{x_2} - \text{proj}_{\mathrm{W}_1}\boldsymbol{x_2} = \boldsymbol{x_2} - \frac{\boldsymbol{x_2}\cdot\boldsymbol{v_1}}{\boldsymbol{v_1}\cdot\boldsymbol{v_1}}\boldsymbol{v_1}$$

Step 3: Now enlarge W_2: let $W_3 = \text{Span}\,\{\boldsymbol{x_1}, \boldsymbol{x_2}, \boldsymbol{x_3}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{x_3}\}$.
Decompose $\boldsymbol{x_3} = \widehat{\boldsymbol{x}}_{\boldsymbol{3}} + \boldsymbol{v_3}$, where $\widehat{\boldsymbol{x}}_{\boldsymbol{3}} = \text{proj}_{W_2}\boldsymbol{x_3}$ and $\boldsymbol{v_3} \in W_2^{\perp}$.
But we don't need "the whole vector $\boldsymbol{x_3}$" to get W_3According to the Lemma, W_3 $= \text{Span}\,\{\boldsymbol{x_1}, \boldsymbol{x_2}, \boldsymbol{x_3}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{x_3}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$, and, by the Lemma, $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ is an orthogonal set.

Of course, we already know a formula for $\boldsymbol{v_3}$: since $W_2 = \text{Span}\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$,

$$\boldsymbol{v_3} = \boldsymbol{x_3} - \text{proj}_{\mathrm{W}_2}\boldsymbol{x_3} = \boldsymbol{x_3} - \frac{\boldsymbol{x_3}\cdot\boldsymbol{v_1}}{\boldsymbol{v_1}\cdot\boldsymbol{v_1}}\boldsymbol{v_1} - \frac{\boldsymbol{x_3}\cdot\boldsymbol{v_2}}{\boldsymbol{v_2}\cdot\boldsymbol{v_2}}\boldsymbol{v_2}$$

Next: Keep repeating this process as long as you can. Suppose we have arrived at $W_i = \text{Span}\{\boldsymbol{x_1}, ..., \boldsymbol{x_i}\} = \text{Span}\{\boldsymbol{v_1}, ..., \boldsymbol{v_i}\}$, where $\{\boldsymbol{v_1}, ..., \boldsymbol{v_i}\}$ is an orthogonal set.

If $i < p$ (*that is, if there are still some* $\boldsymbol{x_i}$*'s remaining*), then use $\boldsymbol{x_{i+1}}$ to enlarge W_i again: $W_{i+1} = \text{Span}\{\boldsymbol{x_1}, ..., \boldsymbol{x_i}, \boldsymbol{x_{i+1}}\} = \text{Span}\{\boldsymbol{v_1}, ..., \boldsymbol{v_i}, \boldsymbol{x_{i+1}}\}$.
Once again, we don't need "the whole vector $\boldsymbol{x_{i+1}}$"; as before, it's enough to use $\boldsymbol{v_{i+1}} =$ the component of $\boldsymbol{x_{i+1}}$ orthogonal to W_i. By the Lemma, we have $\text{Span}\{\boldsymbol{x_1}, ..., \boldsymbol{x_i}, \boldsymbol{x_{i+1}}\} = \text{Span}\{\boldsymbol{v_1}, ..., \boldsymbol{v_i}, \boldsymbol{x_{i+1}}\} = \text{Span}\{\boldsymbol{v_1}, ..., \boldsymbol{v_i}, \boldsymbol{v_{i+1}}\}$ and $\{\boldsymbol{v_1}, ..., \boldsymbol{v_i}, \boldsymbol{v_{i+1}}\}$ is an orthogonal set.

And, of course, we know a formula for $\boldsymbol{v_{i+1}}$: since $W_i = \text{Span}\{\boldsymbol{v_1}, \boldsymbol{v_2}, ...\boldsymbol{v_i}\}$,

$$\boldsymbol{v_{i+1}} = \boldsymbol{x_{i+1}} - \text{proj}_{W_i}\boldsymbol{x_{i+1}} = \boldsymbol{x_{i+1}} - \frac{\boldsymbol{x_3}\cdot\boldsymbol{v_1}}{\boldsymbol{v_1}\cdot\boldsymbol{v_1}}\boldsymbol{v_1} - \frac{\boldsymbol{x_3}\cdot\boldsymbol{v_2}}{\boldsymbol{v_2}\cdot\boldsymbol{v_2}}\boldsymbol{v_2} - ... - \frac{\boldsymbol{x_{i+1}}\cdot\boldsymbol{v_i}}{\boldsymbol{v_i}\cdot\boldsymbol{v_i}}\boldsymbol{v_i}$$

Finally, we run out of $\boldsymbol{x_i}$'s and arrive at $W_p = \text{Span}\{\boldsymbol{x_1}, ..., \boldsymbol{x_p}\} = \text{Span}\{\boldsymbol{v_1}, ..., \boldsymbol{v_p}\}$, and $\{\boldsymbol{v_1}, ..., \boldsymbol{v_p}\}$ is an orthogonal set that is a basis for the original subspace $W = W_p$.

(*For certain purposes it's useful to remember that <u>at each step</u> in the process, the set* $\{\boldsymbol{x_1}, ..., \boldsymbol{x_i}\}$ *has the same span as the set* $\{\boldsymbol{v_1}, ..., \boldsymbol{v_i}\}$

You'll feel much better about the Gram-Schmidt Process if you understand the preceding material about what's going on in the calculations. However, when you solve problems you might just use a "programmed" recipe for the vectors in the orthogonal basis.

Suppose $\{\boldsymbol{x_1}, ..., \boldsymbol{x_p}\}$ is a basis for the subspace W of $\mathbb{R}^n$. Then $\{\boldsymbol{v_1}, \boldsymbol{v_2}, ..., \boldsymbol{v_p}\}$ is an orthogonal basis for W where

$$\begin{aligned}
\boldsymbol{v_1} &= \boldsymbol{x_1} \\
\boldsymbol{v_2} &= \boldsymbol{x_2} - \text{proj}_{\boldsymbol{v_1}}\boldsymbol{x_2} = \boldsymbol{x_2} - \frac{\boldsymbol{x_2}\cdot\boldsymbol{v_1}}{\boldsymbol{v_1}\cdot\boldsymbol{v_1}}\boldsymbol{v_1} \\
\boldsymbol{v_3} &= \boldsymbol{x_3} - \text{proj}_{\boldsymbol{v_1}}\boldsymbol{x_3} - \text{proj}_{\boldsymbol{v_2}}\boldsymbol{x_3} = \boldsymbol{x_3} - \frac{\boldsymbol{x_3}\cdot\boldsymbol{v_1}}{\boldsymbol{v_1}\cdot\boldsymbol{v_1}}\boldsymbol{v_1} - \frac{\boldsymbol{x_3}\cdot\boldsymbol{v_2}}{\boldsymbol{v_2}\cdot\boldsymbol{v_2}}\boldsymbol{v_2} \\
\boldsymbol{v_4} &= \boldsymbol{x_4} - \text{proj}_{\boldsymbol{v_1}}\boldsymbol{x_4} - \text{proj}_{\boldsymbol{v_2}}\boldsymbol{x_4} - \text{proj}_{\boldsymbol{v_3}}\boldsymbol{x_4} \\
&= \boldsymbol{x_4} - \frac{\boldsymbol{x_4}\cdot\boldsymbol{v_1}}{\boldsymbol{v_1}\cdot\boldsymbol{v_1}}\boldsymbol{v_1} - \frac{\boldsymbol{x_4}\cdot\boldsymbol{v_2}}{\boldsymbol{v_2}\cdot\boldsymbol{v_2}}\boldsymbol{v_2} - \frac{\boldsymbol{x_4}\cdot\boldsymbol{v_3}}{\boldsymbol{v_3}\cdot\boldsymbol{v_3}}\boldsymbol{v_3} \\
&\vdots \\
\boldsymbol{v_p} &= \boldsymbol{x_p} - \text{proj}_{\boldsymbol{v_1}}\boldsymbol{x_p} - \text{proj}_{\boldsymbol{v_2}}\boldsymbol{x_p} - ... \qquad ... - \text{proj}_{\boldsymbol{v_{p-1}}}\boldsymbol{x_p} \\
&= \boldsymbol{x_p} - \frac{\boldsymbol{x_p}\cdot\boldsymbol{v_1}}{\boldsymbol{v_1}\cdot\boldsymbol{v_1}}\boldsymbol{v_1} - \frac{\boldsymbol{x_p}\cdot\boldsymbol{v_2}}{\boldsymbol{v_2}\cdot\boldsymbol{v_2}}\boldsymbol{v_2} - ... \qquad ... - \frac{\boldsymbol{x_p}\cdot\boldsymbol{v_{p-1}}}{\boldsymbol{v_{p-1}}\cdot\boldsymbol{v_{p-1}}}\boldsymbol{v_{p-1}}
\end{aligned}$$

Example Use the Gram-Schmidt Process to find an orthogonal basis for

$$W = \text{Span}\left\{ \begin{bmatrix} 0\\2\\0\\1 \end{bmatrix}, \begin{bmatrix} 1\\0\\0\\1 \end{bmatrix}, \begin{bmatrix} 1\\0\\1\\1 \end{bmatrix} \right\} \text{ and explain some of the details at each step.}$$

$$\begin{matrix} \uparrow & \uparrow & \uparrow \\ \boldsymbol{x_1} & \boldsymbol{x_2} & \boldsymbol{x_3} \end{matrix}$$

You can check that $\boldsymbol{x_1}, \boldsymbol{x_2}, \boldsymbol{x_3}$ are linearly independent and therefore form a basis for W. What would happen if we applied the Gram-Schmidt Process to a linearly dependent set of vectors $\{\boldsymbol{x_1}, \ldots, \boldsymbol{x_p}\}$?

Let $\boldsymbol{v_1} = \boldsymbol{x_1} = \begin{bmatrix} 0\\2\\0\\1 \end{bmatrix}$, and $W_1 = \text{Span}\,\{\boldsymbol{x_1}\} = \text{Span}\,\{\boldsymbol{v_1}\}$.

Let $W_2 = \text{Span}\,\{\boldsymbol{x_1}, \boldsymbol{x_2}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{x_2}\}$.

Using the Lemma, we can replace $\boldsymbol{x_2}$ with $\boldsymbol{v_2}$ $=$ the component of $\boldsymbol{x_2}$ orthogonal to W_1

$= \boldsymbol{x_2} - \text{proj}_{\boldsymbol{v_1}} \boldsymbol{x_2}$

$$\text{proj}_{W_1} \boldsymbol{x_2} = \frac{\boldsymbol{x_2} \cdot \boldsymbol{v_1}}{\boldsymbol{v_1} \cdot \boldsymbol{v_1}} \boldsymbol{v_1} = \frac{\begin{bmatrix} 1\\0\\0\\1 \end{bmatrix} \cdot \begin{bmatrix} 0\\2\\0\\1 \end{bmatrix}}{\begin{bmatrix} 0\\2\\0\\1 \end{bmatrix} \cdot \begin{bmatrix} 0\\2\\0\\1 \end{bmatrix}} \begin{bmatrix} 0\\2\\0\\1 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 0\\2\\0\\1 \end{bmatrix} = \begin{bmatrix} 0\\ \frac{2}{5}\\0\\ \frac{1}{5} \end{bmatrix}, \text{ so}$$

$$\boldsymbol{v_2} = \boldsymbol{x_2} - \text{proj}_{W_1} \boldsymbol{x_2} = \begin{bmatrix} 1\\0\\0\\1 \end{bmatrix} - \begin{bmatrix} 0\\ \frac{2}{5}\\0\\ \frac{1}{5} \end{bmatrix} = \begin{bmatrix} 1\\ -\frac{2}{5}\\0\\ \frac{4}{5} \end{bmatrix}$$

Optional: rescale to avoid working with the fractions in the remaining calculations: multiply by 5 and instead use

$$\boldsymbol{v_2} = 5 \begin{bmatrix} 1\\ -\frac{2}{5}\\0\\ \frac{4}{5} \end{bmatrix} = \begin{bmatrix} 5\\ -2\\0\\ 4 \end{bmatrix} \text{ (\textit{explain why rescaling doesn't “mess anything up.”})}$$

Then $\text{Span}\,\{\boldsymbol{x_1}, \boldsymbol{x_2}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{x_2}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$.

(*cont.*)

Let $W_3 = \text{Span}\,\{\boldsymbol{x_1}, \boldsymbol{x_2}, \boldsymbol{x_3}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{x_3}\}$

Using the Lemma, we can replace $\boldsymbol{x_3}$ with $\boldsymbol{v_3}$ $= \text{the component of } \boldsymbol{x_3} \text{ orthogonal to } W_2$

$$= \boldsymbol{x_3} - \text{proj}_{\,W_2}\boldsymbol{x_3}$$
$$= \boldsymbol{x_3} - \text{proj}_{\,\boldsymbol{v_1}}\boldsymbol{x_3} - \text{proj}_{\,\boldsymbol{v_2}}\boldsymbol{x_3}$$

$$\text{proj}_{\,W_2}\boldsymbol{x_3} = \frac{\boldsymbol{x_3}\cdot\boldsymbol{v_1}}{\boldsymbol{v_1}\cdot\boldsymbol{v_1}}\boldsymbol{v_1} + \frac{\boldsymbol{x_3}\cdot\boldsymbol{v_2}}{\boldsymbol{v_2}\cdot\boldsymbol{v_2}}\boldsymbol{v_2} = \frac{\begin{bmatrix}1\\0\\1\\1\end{bmatrix}\cdot\begin{bmatrix}0\\2\\0\\1\end{bmatrix}}{\begin{bmatrix}0\\2\\0\\1\end{bmatrix}\cdot\begin{bmatrix}0\\2\\0\\1\end{bmatrix}}\begin{bmatrix}0\\2\\0\\1\end{bmatrix} + \frac{\begin{bmatrix}1\\0\\1\\1\end{bmatrix}\cdot\begin{bmatrix}5\\-2\\0\\4\end{bmatrix}}{\begin{bmatrix}5\\-2\\0\\4\end{bmatrix}\cdot\begin{bmatrix}5\\-2\\0\\4\end{bmatrix}}\begin{bmatrix}5\\-2\\0\\4\end{bmatrix}$$

$$= \frac{1}{5}\begin{bmatrix}0\\2\\0\\1\end{bmatrix} + \frac{9}{45}\begin{bmatrix}5\\-2\\0\\4\end{bmatrix} = \begin{bmatrix}1\\0\\0\\1\end{bmatrix},$$

so $\quad \boldsymbol{v_3} = \boldsymbol{x_3} - \text{proj}_{\,W_2}\boldsymbol{x_3} \quad = \begin{bmatrix}1\\0\\1\\1\end{bmatrix} - \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\1\\0\end{bmatrix}$

Then $W_3 = \text{Span}\,\{\boldsymbol{x_1}, \boldsymbol{x_2}, \boldsymbol{x_3}\} = \text{Span}\,\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$, and we have replaced all the $\boldsymbol{x_i}$'s. W_3 is the subspace W we were originally given, and an orthogonal basis for W is

$$\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\} = \left\{\begin{bmatrix}0\\2\\0\\1\end{bmatrix}, \begin{bmatrix}5\\-2\\0\\4\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\end{bmatrix}\right\}.$$